

Solution Architecture

Date	06 July 2026
Team ID	TMID-SL-2026-01
Project Name	SmartLender - Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

High-level architecture mapping the Presentation, Application, and ML pipeline layers of Smart Lender.

Layer	Contents
User Layer	Loan Applicant, Credit Officer / Financial Analyst
Frontend Layer	HTML templates: home.html, predict.html, submit.html
Flask Application Layer	Flask backend (app.py) — routing, form handling (POST), input validation, prediction request processing, resu
Prediction Engine	Loads rdf.pkl (trained XGBoost model) + scale1.pkl (scaler); converts user input to a feature vector; returns Ap
Machine Learning Pipeline	Dataset collection → preprocessing (missing values, encoding, SMOTE balancing, scaling) → train/test split →
Deployment Layer	IBM Cloud hosting; web access through a browser

Component Description Table

Component	Description / Role	Technologies Used
Presentation Layer	Collects applicant details and displays the prediction result	HTML, CSS, Jinja2 templates
Flask Application Layer	Routes requests, validates input, calls the prediction engine, renders Flask	Python, Flask
Prediction Engine	Encodes/scales input the same way as training, then runs model inference	scikit-learn, StandardScaler, XGBoost, pickle
ML Pipeline / Training	Produces the trained model and scaler from historical data	pandas, scikit-learn, imbalanced-learn, XGBoost
Data Store	Holds the source dataset and serialized model artifacts	CSV file, .pkl files
Deployment	Hosts the Flask app for organization-wide access	IBM Cloud